

And in the case of isotropic covariance matrix: $\mathbf{C} = \sigma^2 \mathbf{I}_p$:

$$\frac{1}{N} \text{Tr}(\Sigma \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}]) = \frac{\eta}{h\theta^2} (1 - b_2 + \sigma^2 b_1), \quad \frac{1}{N} \text{Tr}(\Sigma_\beta \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}]) = \frac{\sigma^2}{N} \text{Tr}(\Sigma \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}])$$

A.3 RESOLVENT IDENTITIES

Let $\mathbf{Q}$ be the resolvent matrix defined in equation (6). Denote by $\mathbf{Q}_{-v_i}$ the resolvent matrix obtained from the dataset $\mathbf{V}$ by removing the i^{th} sample v_i , i.e:

$$\mathbf{Q}_{-v_i} = \left(\mathbf{Q}^{-1} - \frac{1}{N} \mathbf{v}_i \mathbf{v}_i^\top \right)^{-1}$$

Then, using lemma A.3, we have that:

$$\mathbf{Q} = \mathbf{Q}_{-v_i} - \frac{\mathbf{Q}_{-v_i} \frac{1}{N} \mathbf{v}_i \mathbf{v}_i^\top \mathbf{Q}_{-v_i}}{1 + \frac{1}{N} \mathbf{v}_i^\top \mathbf{Q}_{-v_i} \mathbf{v}_i},$$

and,

$$\mathbf{Q} \mathbf{x}_i = \frac{\mathbf{Q}_{-v_i} \mathbf{x}_i}{1 + \delta}, \quad \mathbf{Q} \tilde{\mathbf{x}}_i = \frac{\mathbf{Q}_{-v_i} \tilde{\mathbf{x}}_i}{1 + \delta_S}, \quad (8)$$

where:

$$\delta = \frac{1}{N} \text{Tr}(\Sigma \bar{\mathbf{Q}}) = \frac{1}{N} \text{Tr}(\bar{\mathbf{Q}}), \quad \delta_S = (\phi(1 - \varepsilon) + \rho\varepsilon) \frac{1}{N} \text{Tr}(\Sigma_\beta \bar{\mathbf{Q}}) = \frac{\alpha}{N} \text{Tr}(\mathbf{C} \bar{\mathbf{Q}}) \quad (9)$$

Let us recall the expression of $\bar{\mathbf{Q}}$ defined in lemma A.4:

$$\begin{aligned} \bar{\mathbf{Q}} &= \left(\frac{\pi(\boldsymbol{\mu} \boldsymbol{\mu}^\top + \mathbf{I}_p)}{1 + \delta} + \frac{\alpha(1 - \pi)(\boldsymbol{\mu}_\beta \boldsymbol{\mu}_\beta^\top + \mathbf{C})}{1 + \delta_S} + \gamma \mathbf{I}_p \right)^{-1} \\ &= (\mathbf{A} + \mathbf{U} \mathbf{U}^\top)^{-1} \end{aligned}$$

where:

$$\mathbf{A} = \frac{\alpha(1 - \pi)}{1 + \delta_S} \mathbf{C} + \left(\gamma + \frac{\pi}{1 + \delta} \right) \mathbf{I}_p, \quad \mathbf{U} = \left(\sqrt{\frac{\pi}{1 + \delta}} \boldsymbol{\mu}, \sqrt{\frac{\alpha(1 - \pi)}{1 + \delta_S}} \boldsymbol{\mu}_\beta \right) \quad (10)$$

Since $\mathbf{C}$ is symmetric and real valued, then it is diagonalizable, and can be written as:

$$\mathbf{C} = \mathbf{P} \mathbf{D} \mathbf{P}^\top$$

where: $\mathbf{P}^{-1} = \mathbf{P}^\top$ is the matrix containing the eigenvectors of $\mathbf{C}$ in its columns, and $\mathbf{D} = \text{Diag}((d_i)_{i=1}^p)$ the diagonal matrix of the eigenvalues of $\mathbf{C}$. Hence, $\mathbf{A}$ can be written as:

$$\mathbf{A} = \mathbf{P} \Delta \mathbf{P}^\top, \quad \Delta = \text{Diag} \left(\gamma + \frac{\pi}{1 + \delta} + \frac{\alpha(1 - \pi)}{1 + \delta_S} d_i \right)_{i=1}^p \quad (11)$$

And using Woodbury's identity in lemma A.2, we get that:

$$\bar{\mathbf{Q}} = \mathbf{A}^{-1} - \mathbf{A}^{-1} \mathbf{U} (\mathbf{I}_2 + \mathbf{U}^\top \mathbf{A}^{-1} \mathbf{U})^{-1} \mathbf{U}^\top \mathbf{A}^{-1}$$

where: $\mathbf{A}^{-1} = \mathbf{P} \Delta^{-1} \mathbf{P}^\top$ and $\Delta^{-1} = \text{Diag} \left(\frac{1}{\gamma + \frac{\pi}{1 + \delta} + \frac{\alpha(1 - \pi)}{1 + \delta_S} d_i} \right)_{i=1}^p$.

Let $\mathbf{M} = (\mathbf{I}_2 + \mathbf{U}^\top \mathbf{A}^{-1} \mathbf{U})^{-1}$, and denote by $M_{i,j}$ its coordinate in the i^{th} row and j^{th} column. We have that:

$$\begin{aligned} \bar{\mathbf{Q}} &= \mathbf{A}^{-1} - \mathbf{A}^{-1} \mathbf{U} \mathbf{M} \mathbf{U}^\top \mathbf{A}^{-1} \\ &= \mathbf{A}^{-1} - \mathbf{A}^{-1} (\zeta_1 \boldsymbol{\mu} \boldsymbol{\mu}^\top + \zeta_2 \boldsymbol{\mu}_\beta \boldsymbol{\mu}_\beta^\top + \zeta_3 (\boldsymbol{\mu} \boldsymbol{\mu}_\beta^\top + \boldsymbol{\mu}_\beta \boldsymbol{\mu}^\top)) \mathbf{A}^{-1} \end{aligned}$$

where:

$$\zeta_1 = \frac{\pi M_{1,1}}{1+\delta}, \quad \zeta_2 = \frac{\alpha(1-\pi)M_{2,2}}{1+\delta_S}, \quad \zeta_3 = \sqrt{\frac{\alpha\pi(1-\pi)}{(1+\delta)(1+\delta_S)}} M_{1,2}$$

Thus,

$$\bar{\mathbf{Q}} = \mathbf{A}^{-1} - \mathbf{A}^{-1} (\zeta_1 \boldsymbol{\mu} \boldsymbol{\mu}^\top + \zeta_2 \boldsymbol{\mu}_\beta \boldsymbol{\mu}_\beta^\top + \zeta_3 (\boldsymbol{\mu} \boldsymbol{\mu}_\beta^\top + \boldsymbol{\mu}_\beta \boldsymbol{\mu}^\top)) \mathbf{A}^{-1} \quad (12)$$

We can further show that:

$$\begin{aligned} M_{1,1} &= \frac{1}{\det(M^{-1})} \left(1 + \frac{\alpha(1-\pi)}{1+\delta_S} \boldsymbol{\mu}_\beta^\top \mathbf{A}^{-1} \boldsymbol{\mu}_\beta \right) \\ M_{1,2} &= \frac{1}{\det(M^{-1})} \left(-\sqrt{\frac{\alpha\pi(1-\pi)}{(1+\delta)(1+\delta_S)}} \boldsymbol{\mu}^\top \mathbf{A}^{-1} \boldsymbol{\mu}_\beta \right) \\ M_{2,1} &= \frac{1}{\det(M^{-1})} \left(-\sqrt{\frac{\alpha\pi(1-\pi)}{(1+\delta)(1+\delta_S)}} \boldsymbol{\mu}^\top \mathbf{A}^{-1} \boldsymbol{\mu}_\beta \right) \\ M_{2,2} &= \frac{1}{\det(M^{-1})} \left(1 + \frac{\pi}{1+\delta} \boldsymbol{\mu}^\top \mathbf{A}^{-1} \boldsymbol{\mu} \right) \\ \det(M^{-1}) &= \left(1 + \frac{\pi}{1+\delta} \boldsymbol{\mu}^\top \mathbf{A}^{-1} \boldsymbol{\mu} \right) \left(1 + \frac{\alpha(1-\pi)}{1+\delta_S} \boldsymbol{\mu}_\beta^\top \mathbf{A}^{-1} \boldsymbol{\mu}_\beta \right) - \frac{\alpha\pi(1-\pi)}{(1+\delta)(1+\delta_S)} (\boldsymbol{\mu}^\top \mathbf{A}^{-1} \boldsymbol{\mu}_\beta)^2 \end{aligned}$$

Lemma A.7 (Delta). *The parameters δ and δ_S defined in equation 9, are given by the following identities:*

$$\delta = \frac{1}{N} \sum_{i=1}^p \frac{1}{\gamma + \frac{\pi}{1+\delta} + \frac{\alpha(1-\pi)}{1+\delta_S} d_i}, \quad \delta_S = \frac{\alpha}{N} \sum_{i=1}^p \frac{d_i}{\gamma + \frac{\pi}{1+\delta} + \frac{\alpha(1-\pi)}{1+\delta_S} d_i}$$

where: $(d_i)_{i=1}^p$ are the eigenvalues of the covariance matrix $\mathbf{C}$.

Proof. Let: $\mathbf{M} = (\mathbf{I}_2 + \mathbf{U}^\top \mathbf{A}^{-1} \mathbf{U})^{-1}$, and denote by $M_{i,j}$ its coordinate in the i^{th} row and j^{th} column. We have that using the expression of $\bar{\mathbf{Q}}$ in equation 12:

$$\bar{\mathbf{Q}} = \mathbf{A}^{-1} - \mathbf{A}^{-1} (\zeta_1 \boldsymbol{\mu} \boldsymbol{\mu}^\top + \zeta_2 \boldsymbol{\mu}_\beta \boldsymbol{\mu}_\beta^\top + \zeta_3 (\boldsymbol{\mu} \boldsymbol{\mu}_\beta^\top + \boldsymbol{\mu}_\beta \boldsymbol{\mu}^\top)) \mathbf{A}^{-1}$$

Then:

$$\begin{aligned} \delta &= \frac{1}{N} \text{Tr}(\bar{\mathbf{Q}}) \\ &= \frac{1}{N} \text{Tr}(\mathbf{A}^{-1}) - \frac{1}{N} \text{Tr}(\mathbf{A}^{-1} (\zeta_1 \boldsymbol{\mu} \boldsymbol{\mu}^\top + \zeta_2 \boldsymbol{\mu}_\beta \boldsymbol{\mu}_\beta^\top + \zeta_3 (\boldsymbol{\mu} \boldsymbol{\mu}_\beta^\top + \boldsymbol{\mu}_\beta \boldsymbol{\mu}^\top)) \mathbf{A}^{-1}) \end{aligned}$$

We have that, when $N \rightarrow \infty$:

$$\frac{1}{N} \text{Tr}(\mathbf{A}^{-1} \boldsymbol{\mu} \boldsymbol{\mu}^\top \mathbf{A}^{-1}) = \frac{1}{N} \boldsymbol{\mu}^\top (\mathbf{A}^{-1})^2 \boldsymbol{\mu} = \mathcal{O}(N^{-1})$$

since $\|\boldsymbol{\mu}\| = \mathcal{O}(N^{-1})$ by assumption 4.1. The same applies for $\boldsymbol{\mu}_\beta$. Thus:

$$\begin{aligned} \delta &= \frac{1}{N} \text{Tr}(\mathbf{A}^{-1}) - \frac{1}{N} \zeta_1 \boldsymbol{\mu}^\top (\mathbf{A}^{-1})^2 \boldsymbol{\mu} - \frac{1}{N} \zeta_2 \boldsymbol{\mu}_\beta^\top (\mathbf{A}^{-1})^2 \boldsymbol{\mu}_\beta - \frac{2}{N} \zeta_3 \boldsymbol{\mu}^\top (\mathbf{A}^{-1})^2 \boldsymbol{\mu}_\beta \\ &= \frac{1}{N} \text{Tr}(\mathbf{A}^{-1}) + \mathcal{O}(N^{-1}) \\ &= \frac{1}{N} \text{Tr}(\Delta^{-1}) + \mathcal{O}(N^{-1}) \\ &= \frac{1}{N} \sum_{i=1}^p \frac{1}{\gamma + \frac{\pi}{1+\delta} + \frac{\alpha(1-\pi)}{1+\delta_S} d_i} + \mathcal{O}(N^{-1}) \end{aligned}$$

Hence we have the desired result for δ in the regime $N \gg 1$ which we considered in our assumption 4.1.

Similarly for δ_S , we have that:

$$\begin{aligned}
\frac{1}{\alpha} \delta_S &= \frac{1}{n} \text{Tr}(\mathbf{C}\bar{\mathbf{Q}}) \\
&= \frac{1}{N} \text{Tr}(\mathbf{C}\mathbf{A}^{-1}) - \frac{1}{N} \zeta_1 \boldsymbol{\mu}^\top \mathbf{A}^{-1} \mathbf{C} \mathbf{A}^{-1} \boldsymbol{\mu} - \frac{1}{N} \zeta_2 \boldsymbol{\mu}_\beta^\top \mathbf{A}^{-1} \mathbf{C} \mathbf{A}^{-1} \boldsymbol{\mu}_\beta - \frac{2}{N} \zeta_3 \boldsymbol{\mu}^\top \mathbf{A}^{-1} \mathbf{C} \mathbf{A}^{-1} \boldsymbol{\mu}_\beta \\
&= \frac{1}{N} \text{Tr}(\mathbf{C}\mathbf{A}^{-1}) + \mathcal{O}(N^{-1}) \\
&= \frac{1}{N} \text{Tr}(\mathbf{D}\Delta^{-1}) + \mathcal{O}(N^{-1}) \\
&= \frac{1}{N} \sum_{i=1}^p \frac{d_i}{\gamma + \frac{\pi}{1+\delta} + \frac{\alpha(1-\pi)}{1+\delta_S} d_i} + \mathcal{O}(N^{-1})
\end{aligned}$$

Which concludes our proof. $\square$

Now let us compute the trace identities that will be useful in the next sections.

Lemma A.8 (Trace identities). *We have the following trace identities:*

$$\begin{aligned}
\frac{1}{N} \text{Tr}((\Sigma \bar{\mathbf{Q}})^2) &= \frac{1}{N} \sum_{i=1}^p \frac{1}{\left(\gamma + \frac{\pi}{1+\delta} + \frac{\alpha(1-\pi)}{1+\delta_S} d_i\right)^2}, \quad \frac{1}{N} \text{Tr}((\Sigma_\beta \bar{\mathbf{Q}})^2) = \frac{1}{N} \sum_{i=1}^p \left(\frac{d_i}{\gamma + \frac{\pi}{1+\delta} + \frac{\alpha(1-\pi)}{1+\delta_S} d_i} \right)^2, \\
\frac{1}{N} \text{Tr}(\Sigma \bar{\mathbf{Q}} \Sigma_\beta \bar{\mathbf{Q}}) &= \frac{1}{N} \sum_{i=1}^p \frac{d_i}{\left(\gamma + \frac{\pi}{1+\delta} + \frac{\alpha(1-\pi)}{1+\delta_S} d_i\right)^2}
\end{aligned}$$

Proof. We have that:

$$\begin{aligned}
\frac{1}{N} \text{Tr}((\Sigma \bar{\mathbf{Q}})^2) &= \frac{1}{N} \text{Tr}((\boldsymbol{\mu} \boldsymbol{\mu}^\top + \mathbf{I}_p) \bar{\mathbf{Q}} (\boldsymbol{\mu} \boldsymbol{\mu}^\top + \mathbf{I}_p) \bar{\mathbf{Q}}) \\
&= \frac{1}{N} \text{Tr}(\bar{\mathbf{Q}}^2) + \mathcal{O}(N^{-1}) \\
&= \frac{1}{N} \text{Tr}((\mathbf{A}^{-1})^2) + \mathcal{O}(N^{-1}) \\
&= \frac{1}{N} \text{Tr}((\mathbf{P} \Delta^{-1} \mathbf{P}^\top)^2) + \mathcal{O}(N^{-1}) \\
&= \frac{1}{N} \text{Tr}((\Delta^{-1})^2) + \mathcal{O}(N^{-1}) \\
&= \frac{1}{N} \sum_{i=1}^p \frac{1}{\left(\gamma + \frac{\pi}{1+\delta} + \frac{\alpha(1-\pi)}{1+\delta_S} d_i\right)^2} + \mathcal{O}(N^{-1})
\end{aligned}$$

Thus we demonstrated the first identity. For the second one, we have that:

$$\begin{aligned}
\frac{1}{N} \text{Tr}((\Sigma_\beta \bar{\mathbf{Q}})^2) &= \frac{1}{N} \text{Tr}((\boldsymbol{\mu}_\beta \boldsymbol{\mu}_\beta^\top + \mathbf{C}) \bar{\mathbf{Q}} (\boldsymbol{\mu}_\beta \boldsymbol{\mu}_\beta^\top + \mathbf{C}) \bar{\mathbf{Q}}) \\
&= \frac{1}{N} \text{Tr}((\mathbf{C} \bar{\mathbf{Q}})^2) + \mathcal{O}(N^{-1}) \\
&= \frac{1}{N} \text{Tr}((\mathbf{C} \mathbf{A}^{-1})^2) + \mathcal{O}(N^{-1}) \\
&= \frac{1}{N} \text{Tr}((\mathbf{D} \Delta^{-1})^2) + \mathcal{O}(N^{-1}) \\
&= \frac{1}{N} \sum_{i=1}^p \left(\frac{d_i}{\gamma + \frac{\pi}{1+\delta} + \frac{\alpha(1-\pi)}{1+\delta_S} d_i} \right)^2 + \mathcal{O}(N^{-1})
\end{aligned}$$

And the same spirit of the proof applies to the last identity. $\square$